

Inventory Balance — Store Apply Bug (Delta vs Physical)

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Khulasa

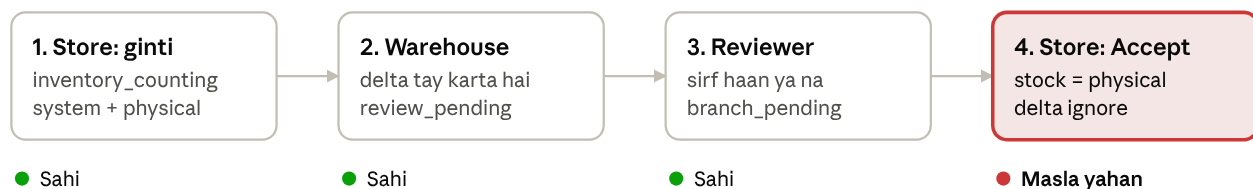
Masla **store (branch) side** par hai: Accept karte waqt store **delta** ko nazar-andaz karke seedha **stock = physical_stock** likh deta hai. Warehouse aur reviewer dono sahi data bhejte hain.

- **Kya ghalat hota hai:** warehouse ka delta override stock par lagta hi nahi, aur ginti ke baad hui sales ki wajah se stock **zyada** dikhne lagta hai.
- **Kahan:** do jagah, aur dono ek hi galat formula use karti hain: store ka local function `applyLocalStockCount()` aur Supabase RPC `apply_inventory_balance_item`.
- **Kab se:** 15 Sep 2026 ki migration `2026_09_15_inventory_balance_apply_absolute.sql`, jis ne delta wala formula hata kar absolute lagaya.
- **Hal:** dono jagah wapas **delta** se apply karna: `naya stock = abhi ka live stock + delta`. Live stock ghalat hone wale purane masle ko alag se handle karna (section "Hal" dekhein).

Feature ka flow

Data sahi chalta hai; bigadta sirf store ke Accept par

Inventory Balance: ginti se stock update tak, 4 qadam



Inventory Balance flow · 4 qadam, masla aakhri qadam par

Store ginti karta hai, warehouse request banata hai, reviewer manzoor karta hai, aur store Accept par stock badalta hai. Pehle teen qadam number sahi aage bhejte hain; chautha qadam `delta` chhod kar ginti seedha stock mein likh deta hai.

Status	Kab	Kaun
<code>review_pending</code>	Warehouse ne request bheji	Warehouse
<code>branch_pending</code>	Reviewer ne Accept kiya	Reviewer
<code>reviewer_rejected</code>	Reviewer ne Reject kiya (khatam)	Reviewer
<code>applied</code>	Store ne Accept kiya, stock badla (khatam)	Store
<code>branch_rejected</code>	Store ne Reject kiya (khatam)	Store

Warehouse side: data sahi nikalta hai

Warehouse ka kaam sirf **request banana** hai; stock badalna us ka kaam nahi. Woh har item ke saath teen number save karta hai, aur teeno sahi hote hain.

Kaam kaise hota hai (Inventory Balance → Create Request tab):

1. Store select hota hai. App Supabase ki `inventory_counting` table se us store ki **last 7 din** ki ginti laata hai, saath `warehouse_products` se naam, SKU aur price.
2. Jo ginti pehle kisi batch mein ja chuki hai, woh list se hat jati hai (`getConsumedCountingIds`), taake ek ginti do dafa na jaye.
3. Har item card par teen number dikhte hain:
 - **SYSTEM** = ginti ke waqt system ka stock (`product_stock`), read-only
 - **PHYSICAL** = store ne jo gina (`counting_stock`), read-only
 - **DELTA** = `PHYSICAL - SYSTEM`. Warehouse isay **stepper se badal sakta hai** (minus bhi), agar use lage ke ginti mein galti hai
4. **Send to Reviewer** dabane par ek batch **warehouse ke local DB** mein ek hi transaction (`runTx`) mein banta hai: batch header + har item (`status = review_pending`) + har item ka `created` log.
5. Background sync yeh rows Supabase par bhej deti hai (`inventory_balance_batches` → `items` → `log`).

Har item mein kya save hota hai:

Column	Matlab	Misaal
<code>system_stock_at_count</code>	Ginti ke waqt system stock	10
<code>physical_stock</code>	Store ki ginti (badalti nahi)	7
<code>delta</code>	Warehouse ka faisla: kitna adjust karna hai	-3 (ya override, jaise -1)
<code>unit_price</code>	Selling price	Rs 200
<code>is_high_variance</code>	`	$\text{delta} \times \text{price}$

Kyun theek hai: `submit()` mein `physical = row.physicalStock` (wahi jo store ne gina) aur `delta = override ?? (physical - system)`. Supabase par yahi values pohanchti hain. Warehouse ka code:

`lib/features/warehouse/inventory_balance/presentation/provider/inventory_balance_provider.dart` → `CreateBatchNotifier.submit()`.

Reviewer (accountant) side: data sahi nikalta hai

Reviewer sirf **haan ya na** karta hai. Woh stock ke kisi number (`physical_stock` , `delta` , `system_stock_at_count`) ko nahi chhoota, is liye warehouse ka data jaisa aaya waisa hi aage jata hai.

Action	Status	Kya update hota hai
Accept	<code>review_pending</code> → <code>branch_pending</code>	<code>reviewer_id</code> , <code>reviewer_name</code> , <code>reviewer_reviewed_at</code> + log <code>reviewer_accepted</code>
Reject	<code>review_pending</code> → <code>reviewer_rejected</code>	Upar wale fields + <code>reviewer_reason</code> + log <code>reviewer_rejected</code>

- Dono update `.eq('status', 'review_pending')` se guard hain, yaani ek item par dobara action nahi hota.
- Reviewer seedha Supabase par likhta hai; warehouse ke local DB mein status `review_pending` hi rehta hai, aur warehouse ka History tab Supabase se padhta hai.

Reviewer ka code:

```
lib/features/accountant/accountant_inventory_review/data/datasource/inventory_review_remote_datasource.dart → acceptItem() / rejectItem().
```

Masla: store side, do jagah

Store jab item **Accept** karta hai to stock do jagah badalta hai, aur **dono jagah** formula `stock = physical_stock` hai. `delta` kahin istemal nahi hota.

Accept ka flow (`branch_inventory_balance_provider.dart` → `acceptItem()`):

1. Pehle Supabase RPC chalta hai (status `applied` + Supabase ka stock).
2. Phir store ke **local DB** mein stock set hota hai. POS aur sale screens yahi local stock dekhti hain.
3. Step 2 fail ho to Supabase item wapas `branch_pending` kar diya jata hai (`revertAppliedItem`), taake Accept dobara ho sake.

Jagah 1: Supabase RPC

- **Function:** `public.apply_inventory_balance_item(p_item_id, p_branch_user_id, p_branch_user_name)`
- **File:** `supabase/migrations/2026_09_15_inventory_balance_apply_absolute.sql`, step 3
- **Abhi:** `v_after := v_item.physical_stock;`
- **Pehle tha** (`2026_09_13_inventory_balance_branch_apply.sql`): `v_after := v_current + v_item.delta;`

Jagah 2: Store ka local DB (asal operational stock)

- **File:** `lib/features/branch/inventory_balance/data/datasource/branch_inventory_balance_datasource.dart`
- **Function:** `applyLocalStockCount({storeId, productId, physicalStock})`
- **Abhi:** `UPDATE branch_stock_inventory SET stock = $1 jahan $1 = item.physicalStock`
- Provider ise `physicalStock: item.physicalStock` deta hai; `item.delta` pass hi nahi hota.

Formula kyun badla gaya tha

15 Sep ki migration ke comment ke mutabiq: live stock pehle hi ghalat (minus) tha, us par purana delta jodne se bakwas number bante the. Misaal: live **-5**, delta **-3**, nateeja **-8**, jabke gina **3** tha. Is liye seedha ginti likhne ka faisla hua.

Yeh **asal bug ka ilaaj nahi tha**. Asal masla ghalat live stock tha; absolute formula ne us ko to chhupa diya, lekin do naye masle khade kar diye (agla section).

Misaalein: absolute formula kya bigaadta hai

Masla A: warehouse ka delta override bekaar ho jata hai

System stock **10**, store ne gine **7**, delta **-3**. Warehouse ko lagta hai 2 item galti se nahi gine gaye, is liye woh stepper se delta **-1** kar deta hai.

	Warehouse ki umeed	Asal mein hota hai
Formula	$10 + (-1)$	<code>stock = physical_stock</code>
Naya stock	9	7

Delta **-1** sirf record aur report mein jata hai (Rs impact, red flag). Stock par asar zero hai.

Masla B: ginti ke baad ki sales nazar-andaz

Ginti, reviewer ka Accept aur store ka Accept, in mein kai din lag sakte hain. Is dauran sales chalti rehti hain.

Din	Waqia	System stock	Asal maal
Somwar	Ginti: system 10, gine 7 (delta -3)	10	7
Budh	2 bik gaye	8	5
Jumma	Store Accept karta hai	?	5

Formula	Nateeja	Sahi?
Absolute (abhi): <code>stock = 7</code>	7	Nahi: 2 item jo mojud hi nahi

Formula	Nateeja	Sahi?
Delta: $8 + (-3)$	5	Haan

Jitni der Accept mein lagti hai, utni sales ki ghalti stock mein aa jati hai, aur hamesha **zyada** stock ki taraf: maal kam hota hai, system zyada batata hai.

Dono formulon ka muqabla

Soorat	Absolute (abhi)	Delta
Ginti ke baad sales ya transfer hon	Ghalat	Sahi
Warehouse ka delta override	Nazar-andaz	Lagta hai
Live stock pehle se ghalat ho	Ginti ka number aa jata hai	Ghalti ke upar delta lagta hai
Ginti aur Accept ke beech kuch na hile	Sahi	Sahi

Hal: dono jagah wapas delta se apply

Tajweez: **Option 1** abhi lagayein. Warehouse aur reviewer ka code nahi badalna; sirf store ka ek Dart function, provider ki ek line aur Supabase ka ek RPC.

Option	Formula	Faida	Nuqsan
1. Delta (tajweez)	$\text{naya stock} = \text{live stock} + \text{delta}$	Sales sahi, warehouse override lagta hai, chhota change	Live stock pehle se ghalat ho to ghalti rehti hai
2. Absolute (abhi)	$\text{naya stock} = \text{physical_stock}$	Ghalat live stock theek ho jata hai	Sales nazar-andaz, override bekaar
3. Ginti + baad ki movement	$\text{naya stock} = \text{physical} + (\text{ginti ke baad ki movement})$	Sab se sahi	Bada kaam: ginti ke waqt se ab tak ki har sale/transfer ka hisaab chahiye

Step 1: Supabase RPC wapas delta par

Nayi migration file, jaise

`supabase/migrations/2026_10_xx_inventory_balance_apply_delta.sql` . Function wahi hai jo 15 Sep wali file mein hai; sirf step 3 badalna hai:

```
-- 3. DELTA apply – live stock + warehouse ka manzoor kiya hua delta
v_after := v_current + v_item.delta;
```

Baaqi function (lock, guard, `applied_stock_before/after` , log) waisa hi rahe. Pehle Supabase **testing**, phir production.

Step 2: Store ka local function

`branch_inventory_balance_datasource.dart` mein `applyLocalStockCount` ki jagah delta wala function:

```
Future<void> applyLocalStockDelta({
  required String storeId,
  required String productId,
  required double delta,
}) async {
  final conn = await DataBaseService.getConnection();
  final result = await conn.execute(
    r'''UPDATE public.branch_stock_inventory
      SET stock = stock + $1, updated_at = NOW()
      WHERE store_id = $2 AND product_id = $3
      RETURNING stock''',
    parameters: [delta, storeId, productId],
  );
  if (result.isEmpty) {
    throw StateError('local_stock_row_not_found');
  }
}
```

`stock = stock + $1` ek hi SQL statement hai, is liye beech mein koi sale ho to bhi hisaab nahi bigadta.

Step 3: Provider mein delta pass karein

`branch_inventory_balance_provider.dart` → `acceptItem()` :

```
await BranchInventoryBalanceRepository.instance.applyLocalStockDelta(
  storeId: item.storeId,
  productId: item.productId,
  delta: item.delta,
);
```

Repository (`branch_inventory_balance_repository.dart`) mein bhi naam aur param isi hisaab se badlein.

Dhyan rakhne wali baatein

- **Delta do dafa nahi lagna chahiye.** Absolute dobara chale to nateeja wahi rehta hai, delta dobara chale to stock dobara badalta hai. Abhi ki tarteeb sahi hai: pehle RPC (sirf `branch_pending` par chalta hai), kamyab ho to local apply, local fail ho to revert. Is tarteeb ko na badlein, aur `applied` item par local apply dobara kabhi na chalayein.
- **Dono jagah ek hi formula.** Store ki sync local se Supabase jati hai, is liye agar RPC aur local alag formule use karein to Supabase ka stock agli sync par badal jayega.
- **Ghalat live stock (minus) ka asal masla** alag hai. Usay is fix mein mix na karein; Accept screen par `stock` pehle → baad dikhana aur minus nateeje par warning dena kaafi hai.
- **Purane `applied` items** ko dobara na chhedein. Jo pehle absolute se lag chuke, unka record `applied_stock_before/after` mein mojud hai.

Testing checklist (fix ke baad, pehle testing DB par)

- ☐ **Seedha case:** system 10, gine 7, delta -3, beech mein koi sale nahi. Accept ke baad store stock **7**.
- ☐ **Sale ke baad:** ginti ke baad 2 bechein (live 8). Accept ke baad stock **5** ($8 + (-3)$), 7 nahi.
- ☐ **Warehouse override:** delta -3 ki jagah -1 bhejein. Accept ke baad stock **9**.
- ☐ **Extra stock:** gine 12 (system 10), delta +2. Accept ke baad stock **12**.
- ☐ **Local fail:** local update fail karwayein (jaise product row na ho). Item wapas `branch_pending` aaye, stock na badle.
- ☐ **Double Accept:** ek item par do dafa Accept. Delta sirf **ek dafa** lage (doosri dafa `item_not_branch_pending`).
- ☐ **Supabase vs local:** agli sync ke baad Supabase ka `branch_stock_inventory.stock` store ke local stock ke barabar ho.

- ☐ **Record:** `inventory_balance_items` mein `applied_stock_before` = live stock aur `applied_stock_after` = live + delta.

Files reference

Side	File	Kaam	Change chahiye?
Store	<code>lib/features/branch/inventory_balance/data/datasource/branch_inventory_balance_datasource.dart</code>	<code>applyLocalStockCount()</code> : local stock set	Haan (delta function)
Store	<code>lib/features/branch/inventory_balance/presentation/provider/branch_inventory_balance_provider.dart</code>	<code>acceptItem()</code> : RPC + local apply + revert	Haan (<code>item.delta</code> pass)
Store	<code>lib/features/branch/inventory_balance/data/repository/branch_inventory_balance_repository.dart</code>	Datasource wrapper	Haan (naam/param)
Supabase	<code>supabase/migrations/2026_09_15_inventory_balance_apply_absolute.sql</code>	RPC ka abhi wala (absolute) version	Nayi migration se replace
Supabase	<code>supabase/migrations/2026_09_13_inventory_balance_branch_apply.sql</code>	RPC ka purana (delta) version	Nahi, sirf reference
Warehouse	<code>lib/features/warehouse/inventory_balance/presentation/provider/inventory_balance_provider.dart</code>	<code>submit()</code> : batch + delta	Nahi
Warehouse	<code>lib/features/warehouse/inventory_balance/data/datasource/inventory_balance_local_datasource.dart</code>	<code>createBatch()</code> : local insert	Nahi
Reviewer	<code>lib/features/accountant/accountant_inventory_review/data/datasource/inventory_review_remote_datasource.dart</code>	Accept / Reject	Nahi